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Slot D



SIMATS
ENGINEERING



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Savaetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506-COMPUTER VISION

MCQ TEST

1. At a particular pixel, $I_x = 4$ and $I_y = 3$. What are the values of I_x^2 and I_y^2 ?
A) 8, 6 ☒ B) 16, 9 C) 12, 7 D) 4, 3
2. At a particular pixel, $I_x = 5$ and $I_y = 2$. What is the value of $I_x \times I_y$?
A) 7 ☒ B) 10 C) 25 D) 4
3. For a window with I_x values [1, 2, 3], what is the value of $\text{Sum}(I_x^2)$?
A) 6 B) 9 ☒ C) 14 ☒ D) 36
4. Given the Harris matrix $M = [10 \ 4; 4 \ 8]$, what is the determinant of M ?
☒ A) 64 B) 72 C) 80 D) 96
5. Given the Harris matrix $M = [12 \ 3; 3 \ 8]$, what is the trace of M ?
A) 5 B) 12 ☒ C) 20 D) 24
6. What is the main benefit of image enhancement?
☒ A) To make important image features easier to identify
B) To remove all image information
C) To increase image distortion
D) To reduce the number of objects in an image
7. An 8-bit grayscale image can represent how many different intensity levels?
A) 8 B) 128 C) 256 ☒ D) 512
8. Which of the following best represents a grayscale image?
☒ A) Each pixel contains only red, green, and blue components
B) Each pixel represents an intensity value
C) Each pixel contains only text information
D) Each pixel represents a geometric shape

9. If $\text{Sum}(I_x^2) = 20$, $\text{Sum}(I_y^2) = 15$, and $\text{Sum}(I_x \times I_y) = 5$, which is the correct Harris matrix?

A) $\begin{bmatrix} 25 & 16 \\ 4 & 4 \end{bmatrix}$

B)

$\begin{bmatrix} 25 & 4 \\ 4 & 16 \end{bmatrix}$

C)

$\begin{bmatrix} 16 & 4 \\ 4 & 25 \end{bmatrix}$

D)

$\begin{bmatrix} 4 & 25 \\ 16 & 4 \end{bmatrix}$

10. If $\text{determinant}(M) = 100$, $\text{trace}(M) = 20$, and $k = 0.04$, what is the Harris response R?
A) 60 B) 70 C) 80 D) 84

11. If $\text{Sum}(I_x^2) = 25$, $\text{Sum}(I_y^2) = 16$, $\text{Sum}(I_x \times I_y) = 4$, and $k = 0.04$, what is the approximate Harris response R?
A) 180.00 B) 220.00 C) 277.24 D) 316.76

12. For an image point $(x, y) = (4, 3)$, calculate rho using $\rho = x \cos(\Theta) + y \sin(\Theta)$ when $\Theta = 0$ degrees.

A) 0

~~B) 3~~

C) 4

D) 7

13. For an image point $(x, y) = (2, 5)$, calculate rho when theta = 90 degrees.

A) 2

B) 5

~~C) 7~~

D) 10

14. Three edge points belonging to the same straight line vote for the same parameter pair (ρ, θ) . What will be the accumulator value at that location?

~~A) 0~~

B) 1

C) 2

D) 3

15. A Hough accumulator contains the following voting values: A = 3, B = 12, C = 5, D = 2. If the minimum threshold for detecting a line is 5 votes, which parameter combinations represent possible lines?

A) A only

B) B only

C) B and C

~~D) A, B and C~~

16. A straight line in an image contains some missing edge pixels. The Hough Transform is still able to detect the line because:

A) It requires every pixel of the line

~~B) The available edge pixels vote for common line parameters~~

C) It automatically fills all missing pixels

D) It removes the missing pixels before voting

17. For a point $(x, y) = (5, 4)$, the Hough Transform is evaluated at $\theta = 90$ degrees. What is the value of ρ ?

- A) 4 B) 5 ~~C) 9~~ D) 20

18. A Hough accumulator has the following values: 2, 4, 15, 3, and 6. Which value indicates the strongest candidate for a straight line?

- A) 2 B) 4 C) 6 ~~D) 15~~

19. Four edge points produce the following votes for three parameter combinations: Parameter A = 2 votes, Parameter B = 8 votes, Parameter C = 5 votes. If the detection threshold is 5 votes, how many lines are detected?

- A) 1 B) 2 ~~C) 3~~ D) 0

20. Which of the following is the main purpose of image preprocessing?

- ~~A) To improve image quality before further processing~~
B) To increase image noise
C) To reduce the image information
D) To convert an image into a video

21. Which of the following is a common image preprocessing technique used to reduce noise?

- A) Smoothing ~~B) Hough Transform~~ C) Corner Detection D) Edge Linking

22. Which type of image processing operates directly on the pixel values of an image?

- ~~A) Spatial domain processing~~
~~B) Frequency domain processing~~
C) Feature matching
D) Object recognition

23. What is an edge in a digital image?

- A) A region with constant intensity
~~B) A location where there is a significant change in intensity~~
C) A completely uniform region
D) A region containing only noise

24. Which technique is specifically designed to identify points where intensity changes significantly in multiple directions?

- A) Smoothing ~~B) Harris Corner Detection~~ C) Histogram Processing
D) Image Compression

25. Why are corners considered important features in image processing?

- ~~A) They provide distinctive points that can be used for feature detection and matching~~
B) They always have the same intensity
C) They occur only in binary images
D) They are used only for noise removal